

PC-03 · EMULSIFIERS & THICKENERS · PERSONAL CARE

HS Span 80

Sorbitan Oleate

INCI	CAS	EC	FAMILY	SUPPLIED AS
Sorbitan Oleate	1338-43-8	215-665-4	PC-03	Oily liquid

01 POSITIONING

Sixty years of emulsion literature assume this molecule is on your bench.

02 HOW IT WORKS

- 01 Low-HLB anchor
Oil-soluble head anchors the interfacial film from the oil side (HLB ~4.3).
- 02 The classic pair
With polysorbates, forms the most documented emulsifier system in the industry.
- 03 W/O capability
Enables water-in-oil systems: barrier creams, sunscreens, ointments.

03 APPLICATIONS

- W/O creams and ointments
- Emulsifier pairs with polysorbates
- Metalworking and lubricant emulsions
- Explosive emulsions (industrial)
- Crop-protection formulations

04 FORMULATION GUIDE

USE LEVEL1–5%

PH & CONDITIONSStable pH 4–9

- Dissolve in the oil phase; combine with a high-HLB partner to hit the oil's required HLB.
- For W/O, keep internal water phase below ~40% and add slowly under homogenization.

05 TYPICAL SPECIFICATION

AppearanceAmber oily liquid

Saponification value145–160

HLB~4.3

Acid value≤ 8

Typical parameters. The specification on the current TDS and the per-lot Certificate of Analysis govern.

06 PERFORMANCE – DIRECTIONAL

W/O emulsion stability

Span 80 systems

Documented use history

Sorbitan ester class

Values are representative of the ingredient class, based on published technical literature, and shown directionally for comparison. Final performance must be validated in the customer's formulation.